Compatibility resolver

Select matching semantics. Fail before effects.

ELI16 · THE IDEA IN PLAIN LANGUAGE

An adapter checklist: matching the plug shape is not enough; voltage, capacity and required guarantees must match too.

Why it matters: Provider substitution is safe only when consumers state requirements independently and mismatches cannot quietly weaken behavior.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	**Resolver**
		Inference · C4	Authorization

Pure deterministic filtering of descriptors against explicit version, profile, operation, guarantee and resource requirements.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

No mutable domain state

A pure selection function; callers retain chosen descriptor identity

Compatibility decision

Filter/error ordering and deterministic tie-break

OUTSIDE THIS RESPONSIBILITY

Execution or liveness

Selecting a descriptor does not prove the provider is reachable or correct

Evidence verification

Required verification currently fails; no verifier is implemented

same-major/1 responsibility; explicit exact profiles; M1 wire remains 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
Version filter	Same major and explicit [minimum, maximum) range	No candidate → UNSUPPORTED_VERSION
Profile/features filter	Exact profile and required feature subset	Mismatch → UNSUPPORTED_PROFILE
Operation filter	Required operation subset	Mismatch → UNSUPPORTED_OPERATION
Guarantee/resource filter	Guarantees, payload, privacy and latency requirement	Unsatisfied → NOT_AVAILABLE
Deterministic selection	Highest matching version/provider ordering	No hidden semantic relaxation

Trusted-local metadata checks; no proof of advertised reliability or measured performance.

A worked example through the boundary

EXECUTED LOCAL RESOLUTION · positive/negative descriptors are fixtures

01 · STEP

State need

Consumer requires scoped, idempotent process restart.

02 · STEP

Reject mismatch

Wrong profile, missing operation or weak guarantee fails.

03 · STEP

Choose match

Resolve explicit provider and profile.

04 · STEP

Check again

Binding verifies actual token before an effect.

What the example establishes

Compatibility is a conjunction of requirements, not a matching name.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Independent requirement

Do not derive the required guarantees from whichever provider appears.

I2

Exact semantics

Same major alone is insufficient; profile/features/operations must match.

I3

Fail-closed proof need

An unsupported required verifier cannot become optional.

I4

No silent drift

Changed selected provider/token is refused by the binding.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Provider configuration changes after selection	Invocation rejects descriptor drift.	Caller must select explicitly again
No eligible candidate	Return a typed compatibility error; create no domain effect.	No weaker fallback
Restart with saved intent	Use the pinned selection in that intent.	Fresh replacement needs explicit composition

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

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Ordered compatibility stages

Version → profile → operations → resources → selection

1.2.3

Additive same-major fixture

Resolver test; actual M1 wire stays 1.0.0

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Runtime effects from resolution

Pure-function boundary

These are deterministic conformance properties, not measured selection performance. Advertised latency metadata is not a benchmark or verified SLO from this resolver.

PROPOSED BENCHMARK · NOT RUN

Proposed: increasing descriptor count and rejection-stage mixes; measure p50/p95 selection time and allocations, while preserving exact error and tie-break outcomes.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Executed pure Python resolution and descriptor-bound in-process/stdio calls.
FIXTURE / LOCAL TEST POLICY	Synthetic versions, profiles, guarantees, payload limits and SLO metadata.
BLOCKED / UNSUPPORTED	A requirement for verification cannot pass until a supported verifier exists.
UNRUN	Large dynamic catalogues, remote attestation, availability optimization and adaptive scheduling.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Explain selection decisions	Structured reason trace without relaxing filters	Operators can reproduce each rejection
Connect verified evidence	Explicit verifier and freshness contract	Only independently checked claims satisfy proof requirements
Measure catalogue scale	Controlled sizes and worst-case rejection mix	Deterministic semantics survive optimization

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Compatibility is a conjunction of requirements, not a matching name.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py --fixture-only
```

Positive and negative descriptor fixtures are the appropriate evidence here; they do not validate a provider’s real-world SLO or security.

`fusion/registry.py`
`fusion/bindings.py`
`tests/test_contract.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.